Uncertainty-aware damage identification in short-span bridges via physics-informed variational autoencoder

Revision notes: case study, datasets, corrections of the SciML model and validation

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Revision of MS CPENG-8674 — October 29, 2026

INTRODUCTION

The reviewers raised three main concerns about the numerical study of the submitted manuscript: the structural model was too simple (a five-element, simply supported beam), the test data came from the same model used for training, and measurement noise was the only source of uncertainty. In addition, a detailed inspection of the code revealed implementation errors that affected the estimated posteriors and the reference (ground-truth) posteriors used to assess them. This document describes how we addressed these issues. We keep the methodology of the submitted version (encoder, Gaussian copula sampling layer, differentiable eigenvalue decoder, and the loss of Eq. (10)) and follow its notation.

The document has four parts. The first describes the enhanced case study: a bridge on elastic bearings, a finer finite element (FE) model to generate the test data, and the sources of uncertainty that the decoder does not represent. The second describes the datasets and explains how these uncertainty sources enter the inference. The third describes the corrections of the implementation. The fourth describes how we compute the reference posteriors and compare them with the VAE estimates. Table 1 relates each change to the reviewers' comments; the code is available in `MODULES/review29oct_fixed`.

TABLE 1: Correspondence between the changes and the reviewers' comments.

Comment	Change
R1.1, R1.7	The test observations come from a 50-element FE model with the same damage zones and sensors as the 10-element model embedded in the decoder.
R1.2, R3.1	Temperature, mass and within-zone stiffness variability are present in all simulated data; their effect is included in the likelihood as a calibrated model discrepancy.
R1.3	Coverage is evaluated at several nominal levels using the exact marginal distributions of the VAE.
R1.6	The training set is balanced across healthy, mildly damaged and localized damage scenarios; we report the false-alarm rate.
R2.2, R2.3	The bridge rests on elastic bearings, each with its own stiffness reduction factor.
R3.2	Hyperparameters and the final model are selected on the validation set only.
R3.3	The prior of the ELBO (uniform) is independent of the sampling density used to generate the training set.
R3.4	We quantify how the observed quantities and the number of modes control the width of the posterior.
R3.7	The ELBO is estimated with $H = 6$ samples per observation.

ENHANCED CASE STUDY

Bridge on elastic bearings

Following Reviewer 2, we replace the simply supported beam of Eq. (2) by a beam resting on elastic bearings (Fig. 1). Each bearing is represented by a vertical spring of stiffness k_v and a rotational spring of stiffness k_r. Damage can now occur in the deck and in the bearings. We divide the deck into five zones of equal length and assign one stiffness reduction factor to each zone and one to each bearing. The damage condition vector therefore has seven components,

$$\mathbf{z} = \{z_1, \dots, z_5,\, z_{b,L},\, z_{b,R}\}^\top \in \mathcal{S} = [z_{\mathrm{lb}}, 1]^7, \qquad z_{\mathrm{lb}} = 0.45, \tag{1}$$

which define the seven-dimensional latent space of the VAE. The dimension of the latent space depends only on the number of zones and bearings, not on the number of finite elements: each zone k may contain several elements $e \in \mathcal{Z}_k$, all of them scaled by the same factor z_k. The updated global stiffness matrix of Eq. (5) thus becomes

$$\hat{\mathbf{K}}_{\mathrm{G}}(\mathbf{z}) = \sum_{k=1}^{5} z_k \sum_{e \in \mathcal{Z}_k} \mathbf{K}_{e,\mathrm{G}} + z_{b,L}\, \mathbf{K}_{b,L} + z_{b,R}\, \mathbf{K}_{b,R}, \tag{2}$$

where $\mathbf{K}_{b,L}$ and $\mathbf{K}_{b,R}$ contain the nominal bearing stiffnesses k_{v0} and k_{r0} at the vertical and rotational DOFs of the end nodes. Since the bearings are elastic, no DOF is removed and $n_{\mathrm{dof}} =$

$2(n_{\text{el}}+1)$. The mode shapes are observed at the six zone boundaries by vertical accelerometers, regardless of the mesh. Table 2 summarizes the main specifications.

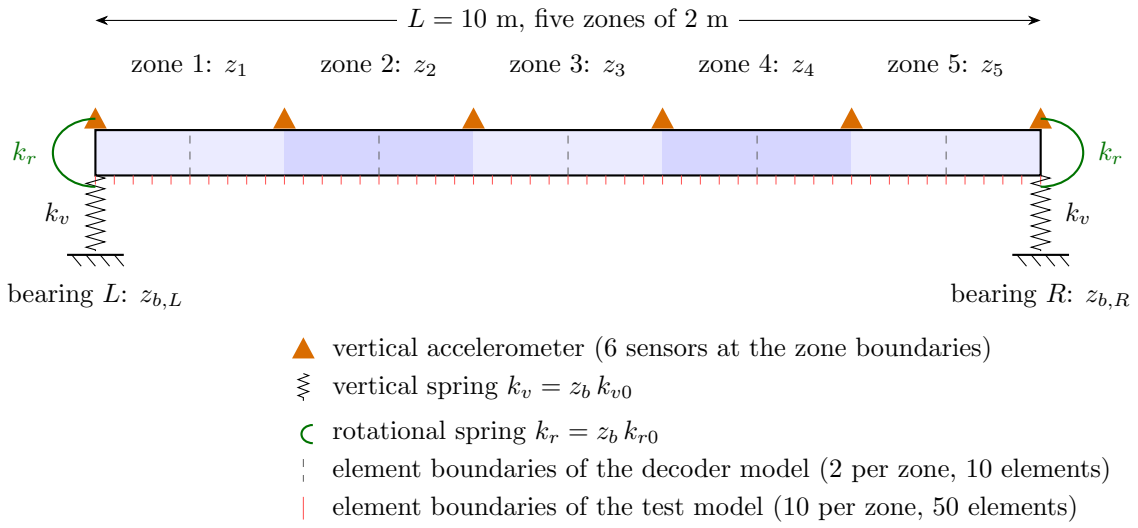


FIG. 1: Bridge on elastic bearings. The seven unknowns are the stiffness reduction factors of the five deck zones ($z_1, \ldots, z_5$) and of the two bearings ($z_{b,L}$, $z_{b,R}$, which scale both springs of each bearing). The VAE decoder uses a 10-element model; the test observations are generated with a 50-element model.

TABLE 2: Main specifications of the case study.

Deck	$L = 10$ m, $E = 210$ GPa, $\rho = 7850$ kg/m^3, $A = 0.005$ m^2, $I = 8.33 \times 10^{-6}$ m^4
Bearings (each end)	$k_{v0} = 1.5 \times 10^8$ N/m, $k_{r0} = 5 \times 10^6$ N m/rad
Unknowns (latent space)	$\mathbf{z} \in [0.45, 1]^7$: five deck zones and two bearings
Decoder model	10 elements (2 per zone), 22 DOFs, nominal mass and modulus
Test model	50 elements (10 per zone), 102 DOFs
Observations	first $n_m = 5$ frequencies and vertical mode shapes at $n_n = 6$ sensors ($d = 35$)

Discretization of the decoder and of the test model

Reviewer 1 asked us to test the trained VAE on observations generated by a finer mesh, keeping the same damage zones and sensors. Such a test is only meaningful if the model embedded in the decoder is accurate enough. With one element per zone, as in the submitted version, the fifth frequency differs by 13% from a converged model (Table 3). This error is much larger than the measurement noise, so the VAE would fail on finer-mesh data for reasons unrelated to the method. With two elements per zone, the error drops below 0.5%. We therefore embed a 10-element model in the decoder and generate the test observations with a 50-element model.

The remaining discrepancy between both models is small and is handled as described in the next section.

TABLE 3: Relative error (%) of the natural frequencies of the healthy bridge with respect to a 200-element model.

Model	f_1	f_2	f_3	f_4	f_5
5 elements (submitted version)	0.04	0.32	1.15	2.42	13.33
10 elements (decoder)	0.003	0.02	0.08	0.22	0.48
50 elements (test model)	0.000	0.000	0.000	0.000	0.001
Frequencies of the 50-element model [Hz]	6.68	18.58	36.65	60.84	91.09

Sources of uncertainty not represented by the decoder

In practice, the measured modal properties of a bridge do not depend only on its stiffness. Reviewers 1 and 3 asked us to consider other sources of variability. We include three of them in every simulated structure:

- *Temperature*, which changes the elastic modulus of the whole deck by the same factor $1+\delta_T$, with $\delta_T \sim \mathcal{N}(0, 0.03^2)$.
- *Mass uncertainty* (non-structural mass, pavement), as a factor $1+\mu_e$ on the mass of each element, with $\mu_e \sim \mathcal{N}(0, 0.02^2)$.
- *Spatial variability of stiffness* within each zone, as a factor $1/(1+\varepsilon_e)$ on each element, with $\varepsilon_e \sim \mathcal{N}(0, 0.025^2)$ and zero mean inside the zone, so that the equivalent stiffness of the zone remains exactly z_k.

The element matrices of a simulated ("true") structure are therefore

$$\mathbf{K}_e^{\text{true}} = \frac{(1+\delta_T)\, z_{k(e)}}{1+\varepsilon_e}\, \mathbf{K}_e, \qquad \mathbf{M}_e^{\text{true}} = (1+\mu_e)\, \mathbf{M}_e. \tag{3}$$

These effects are deliberately *not* included in the decoder, which remains the eigenvalue solver of the nominal 10-element model, as in the submitted methodology. The next section explains how the inference accounts for them.

DATASETS AND LIKELIHOOD

How unmodelled variability enters the inference

The decoder computes the modal properties $\mathbf{g}(\mathbf{z})$ of the nominal model for a given damage condition. The measured modal properties, however, also reflect temperature, mass, mesh refinement

and measurement noise. We write the observations as

$$\mathbf{m} = \mathbf{g}(\mathbf{z}) + \underbrace{\mathbf{b} + \boldsymbol{\epsilon}_{\text{model}}}_{\text{temperature, mass, mesh}} + \boldsymbol{\epsilon}_{\text{meas}}, \tag{4}$$

where $\mathbf{b}$ is a systematic bias, $\boldsymbol{\epsilon}_{\text{model}}$ a random model discrepancy and $\boldsymbol{\epsilon}_{\text{meas}}$ the measurement noise. The stiffness factors $\mathbf{z}$ are not meant to absorb the discrepancy. If the loss ignored it, a frequency drop caused by a warm day could only be explained by reducing some z_k, producing a false damage detection with an overconfident posterior. Instead, the discrepancy is part of the likelihood: we estimate the statistics of $\mathbf{b} + \boldsymbol{\epsilon}_{\text{model}} + \boldsymbol{\epsilon}_{\text{meas}}$ by simulation and use them to weight the data terms of the loss (Eq. 5). This approach is standard in Bayesian FE model updating and leaves the architecture unchanged.

The calibration of the likelihood has two important consequences. First, the observations are treated as less precise than the measurement noise alone would suggest, which widens the posterior consistently with what the decoder cannot represent. Second, temperature changes all frequencies together, so the frequency errors are strongly correlated across modes. The likelihood therefore recognizes a uniform shift of all frequencies as a plausible temperature effect, whereas a shift with a localized pattern, together with changes in the mode shapes, points to damage.

Calibration of the likelihood

We simulate 6 000 structures over the design of experiments described below. For each of them, we compare the modal properties of the 50-element structure of Eq. (3) with those of the decoder at the same $\mathbf{z}$. For the logarithms of the frequencies, the mean difference gives the bias $\mathbf{b}$, which ranges from -0.05% (mode 1) to -0.53% (mode 5) and is due to the mesh. Their covariance, plus the identification noise, gives the matrix $\boldsymbol{\Sigma}_f$. The resulting standard deviation is 1.6% for every mode, and the correlation between modes is about 0.6, due to temperature. For the vertical mode shapes, the deviation from the decoder shapes (measured through the MAC) gives a standard deviation $s_{v,j}$ per mode between 2.8% and 3.9% of the unit-norm shape.

We verify the calibration on independent test data: at the true damage condition, the average value of each likelihood term must equal its theoretical value. The ratios lie between 0.98 and 1.01 (Fig. 2), which confirms that the likelihood describes the actual deviations between the observations and the decoder.

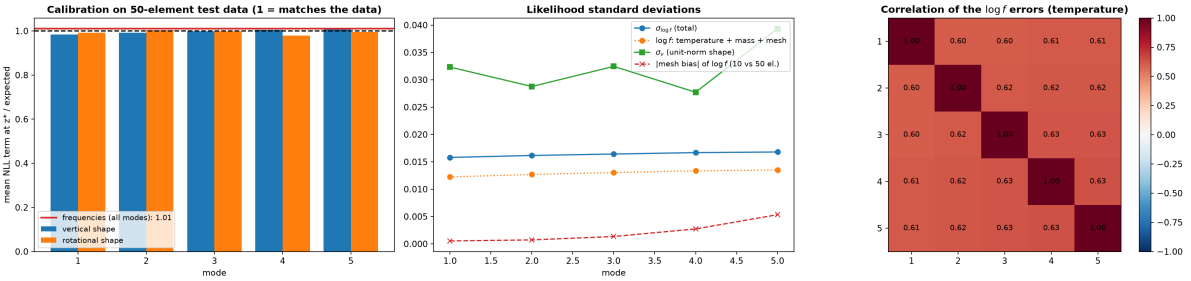


FIG. 2: Calibration of the likelihood against the 50-element test structure. Left: average likelihood terms at the true state divided by their theoretical value (1 means calibrated). Centre: standard deviations of the log-frequencies and of the vertical mode shapes. Right: correlation between the log-frequency errors of different modes, caused by temperature.

Design of experiments

The submitted version sampled every stiffness reduction factor from $\mathcal{U}[0.5, 1]$, so most samples had all elements damaged simultaneously (Reviewer 1, comment 6). The new design of experiments (DoE) represents realistic and balanced scenarios. Each sample belongs to one of the categories of Table 4. Within each category, every damage location (five zones and two bearings), or combination of locations, appears the same number of times. The damaged factors are drawn from $[0.45, 0.95)$ with Latin hypercube stratification, which covers severe, moderate and mild damage uniformly, and the undamaged factors are drawn from $[0.95, 1]$. As a result, every parameter is damaged in 33.2% of the training samples (Fig. 3).

TABLE 4: Scenario categories of the design of experiments.

Category	Share	Damaged parameters
Healthy	10%	none
Single damage	30%	one (each of the seven locations equally often)
Double damage	25%	two (all 21 pairs)
Triple damage	15%	three (all 35 triples)
Multiple damage	10%	four or five (all 56 combinations)
Uniform	10%	$\mathbf{z} \sim \mathcal{U}[0.45, 1]^7$, to cover the whole support of the prior

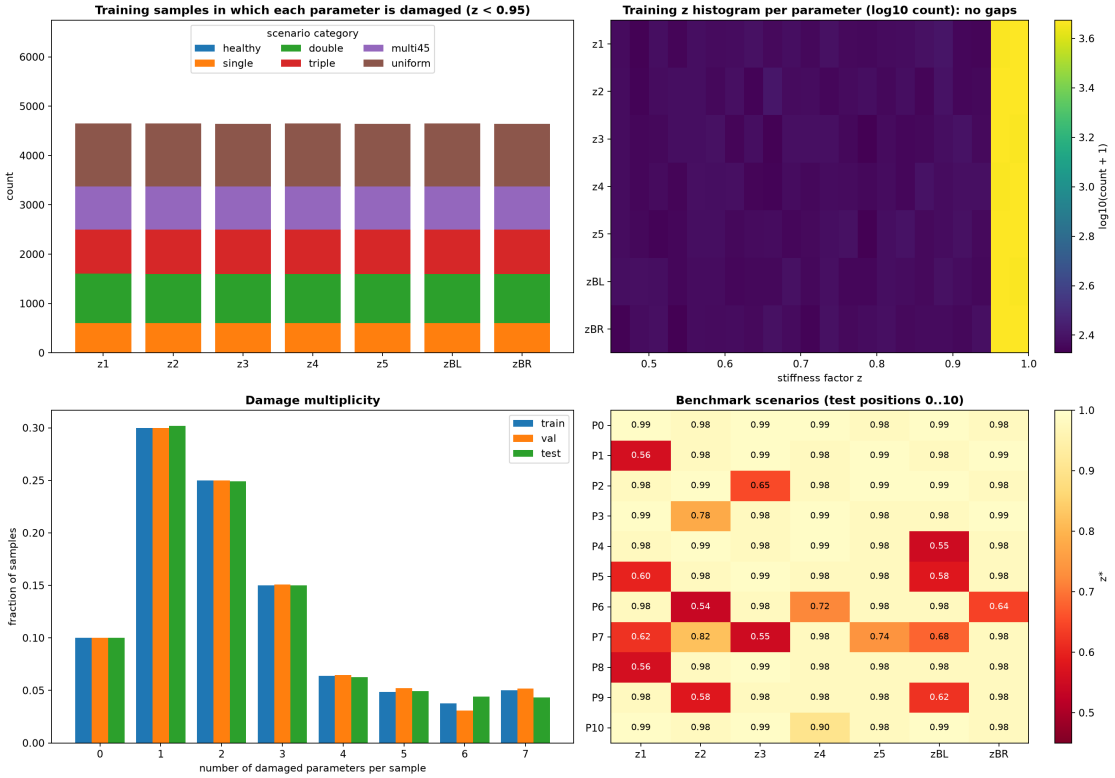


FIG. 3: Design of experiments: number of training samples in which each parameter is damaged, by category; histogram of the stiffness reduction factors; number of damaged parameters per sample; and the eleven benchmark test scenarios.

Training, validation and test datasets

The three datasets differ in their purpose and in the model that generates them (Table 5). The training and validation observations are generated with the 10-element model, and the test observations with the 50-element model, so that the VAE is evaluated on data from a more refined model than the one it was trained with. All of them include temperature, mass and within-zone variability (Eq. 3). We solve the generalized eigenvalue problem of each structure, keep the first five modes, and add measurement noise as in the submitted version: a multiplicative noise of 1% on the frequencies, and an additive noise on the mode shapes with a standard deviation of 5% of the peak amplitude of each mode. The noisy shapes are normalized to unit length and their sign is aligned with the healthy shapes.

We observe the vertical mode shapes only, because accelerometers measure vertical motion; the observation vector is $\mathbf{m} = \text{flat}(\{\mathbf{f}, \mathbf{\Phi}\}) \in \mathbb{R}^{35}$. The first eleven test samples are benchmark scenarios (Fig. 3, bottom right): ten of them keep the damage conditions of the submitted version and one adds a case of mild damage ($z_4 = 0.90$). All of them are plain simulations of a single structure.

TABLE 5: Datasets.

	Training	Validation	Test
Number of samples	14 000	2 400	1 200 (11 benchmarks)
Model that generates the data	10 elements	10 elements	50 elements
Temperature, mass, within-zone variability	yes	yes	yes
Measurement noise (1% frequencies, 5% shapes)	yes	yes	yes
Purpose	minimize the ELBO	select the model	evaluate the model

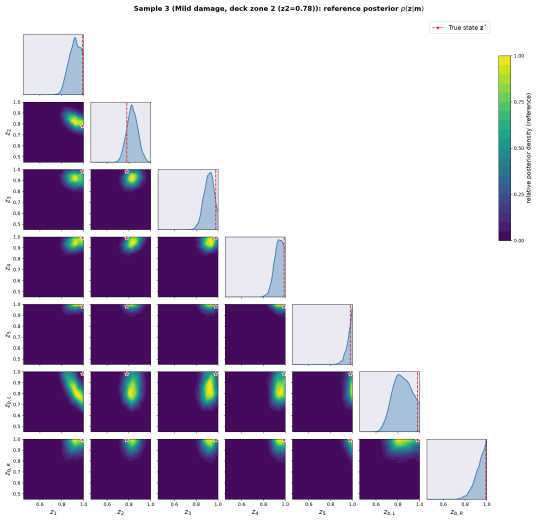
Information content and width of the posterior

The width of the posterior reflects how much information the observations contain. Table 6 shows the standard deviation of the exact posterior for three benchmark scenarios under different measurement setups. When the rotations are also measured at every node with low noise, as in the intermediate version of our code, the posterior is very concentrated and a deterministic estimate would be almost as informative. Observing the vertical mode shapes only, with realistic noise and model discrepancy, yields posteriors with substantial uncertainty, which varies across scenarios and parameters. Using fewer modes widens them further (Reviewer 3, comment 4).

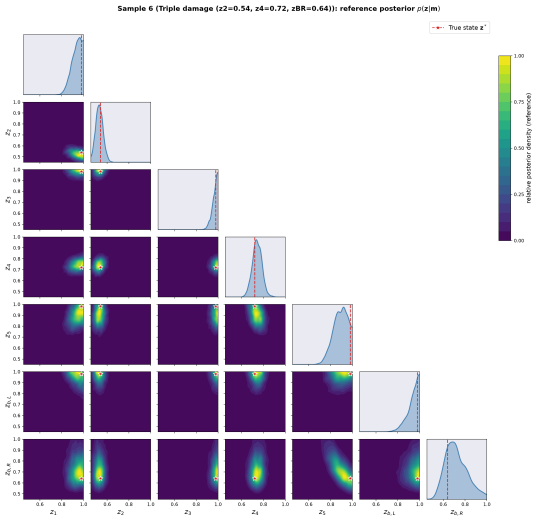
TABLE 6: Standard deviation of the exact posterior marginals (range over the seven parameters) for benchmarks P3, P5 and P9.

Observed quantities and noise (frequencies / shapes)	Posterior standard deviation
Frequencies, vertical and rotational shapes, 1.5% / 3%	0.013–0.028
Frequencies and vertical shapes, 1.5% / 3%	0.016–0.044
Frequencies and vertical shapes, 2.5% / 5%	0.024–0.073
Frequencies and vertical shapes, 2.5% / 5%, three modes	0.031–0.153
Adopted setup (Table 2), all eleven benchmarks	0.02–0.13

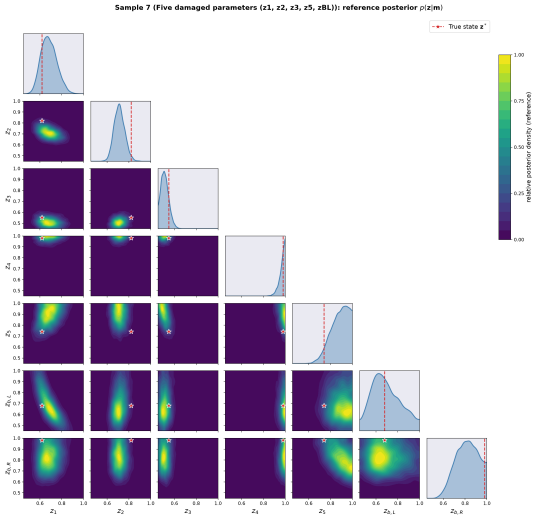
Figure 4 shows the exact posteriors of four benchmark scenarios with the adopted setup. The marginals are asymmetric, the bearings are the most uncertain parameters (95% credible intervals up to 0.47 wide), and several pairs of parameters are clearly correlated. For example, in P9 a stiffer deck near the left support can compensate a weaker left bearing (z_1 and $z_{b,L}$), and in P7 the same happens at the right end (z_5 and $z_{b,R}$). The true damage condition lies within the 95% credible interval for six or seven of the seven parameters in every scenario.



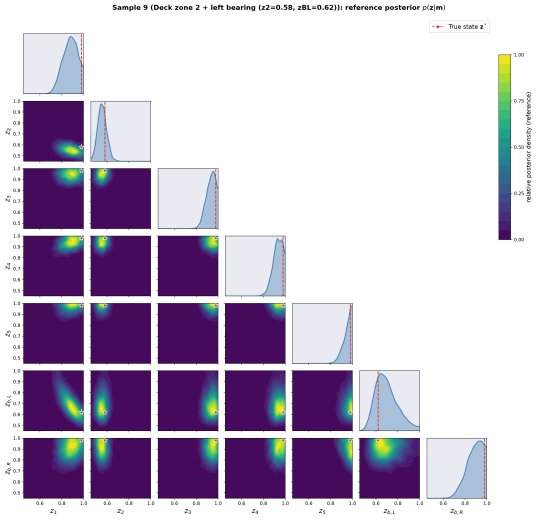
(a) P3: mild damage in zone 2 ($z_2 = 0.78$)



(b) P6: $z_2 = 0.54$, $z_4 = 0.72$, $z_{b,R} = 0.64$



(c) P7: five damaged parameters



(d) P9: zone 2 and left bearing

FIG. 4: Exact posteriors $p(\mathbf{z}|\mathbf{m}_t)$ of four benchmark scenarios, computed from 50-element test observations. The red star marks the true damage condition.

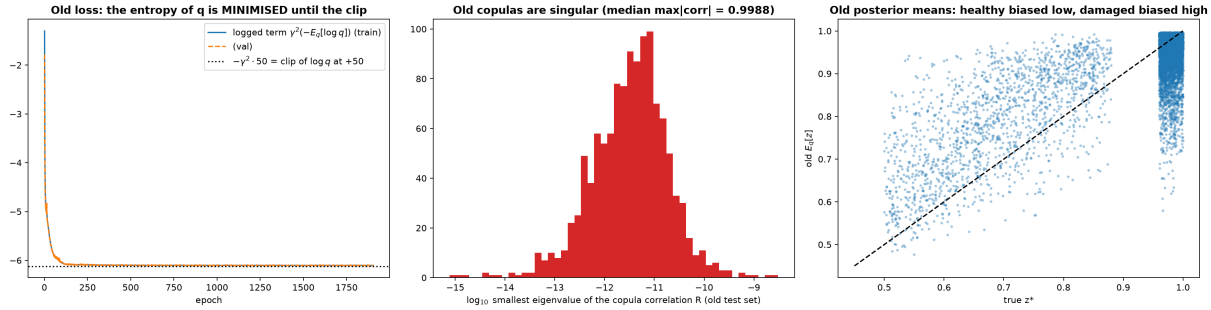
CORRECTIONS OF THE SCIML MODEL

The methodology of the submitted version is preserved. Following the limitation identified in its conclusions, the truncated Gaussian marginals are replaced by mixtures of $K = 3$ Kumaraswamy distributions, which are bounded and can represent multimodal marginals. During the revision we found several errors in the implementation that made the estimated posteriors unreliable. This section describes them, the corrected formulation of the loss, and the verification of the corrected code.

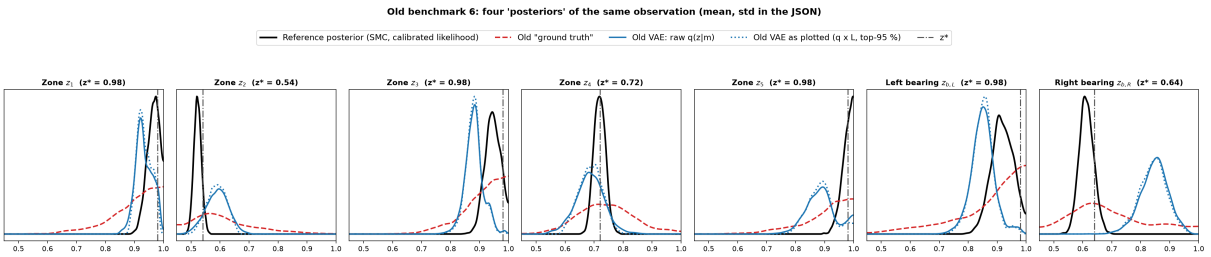
Errors found in the previous implementation

The most important error concerned the sign of the PDF term. The ELBO requires the expected log-density of the approximate posterior, $\mathcal{L}_{\mathrm{PDF}} = \mathbb{E}_q[\log q_{\boldsymbol{\zeta}_{\boldsymbol{\theta}}}(\mathbf{z}|\mathbf{m})]$, which penalizes posteriors that are too concentrated. The code implemented the opposite sign, so minimizing the loss rewarded posteriors that were as concentrated as possible. The only limit was a clipping of $\log q$ at $+50$: the logged PDF term stayed exactly at this limit for most of the training (Fig. 5a), and the encoder reached it by making the copula correlation matrices nearly singular. This explains the thin, ridge-like joint densities observed in the previous results.

Three further issues affected the results. An auxiliary penalty pushed the samples away from $z = 1$, where the posterior of every undamaged parameter is located; consequently, undamaged parameters were underestimated by 0.02 to 0.06, while damaged ones were overestimated by 0.06 to 0.15. The gradients of the Kumaraswamy sampling layer were clipped in several places, and the learning rate decreased by four orders of magnitude during training, which effectively stopped the optimization after a few hundred epochs. Finally, the previous reference posterior was not a valid posterior: part of its evaluation points were drawn around the true damage condition, the corresponding correction weights were only partially applied, and its likelihood used a manually tuned temperature. As a result, it was four to five times wider than the exact posterior (Fig. 5b), which made the VAE estimates look too narrow by comparison.



(a)



(b)

FIG. 5: Diagnosis of the previous implementation. (a) The PDF term remains at its clipping limit, the copula correlation matrices are nearly singular, and the posterior means are biased. (b) Four estimates of the posterior of the same observation (benchmark 6): exact posterior (black), previous reference (red, dashed), previous VAE (blue) and previous VAE as plotted after re-weighting (blue, dotted).

Corrected loss function

The loss keeps the structure of Eq. (10). The difference lies in the weights of the frequency and MAC terms: instead of fixed values, they are now given by the calibrated statistics of the discrepancy between observations and decoder (Eq. 4):

$$\mathcal{L}_{\mathrm{ELBO}}(\boldsymbol{\theta}) = \underbrace{\frac{1}{NH}\sum_{i=1}^{N}\sum_{h=1}^{H}\frac{1}{2}\,\mathbf{d}_{ih}^{\top}\boldsymbol{\Sigma}_f^{-1}\mathbf{d}_{ih}}_{\mathcal{L}_{\mathrm{freq}}} + \underbrace{\frac{1}{NH}\sum_{i=1}^{N}\sum_{h=1}^{H}\sum_{j=1}^{n_m}\frac{1-\mathrm{MAC}_j(\boldsymbol{\Phi}_i,\hat{\boldsymbol{\Phi}}_i^h)}{2s_{v,j}^2}}_{\lambda\mathcal{L}_{\mathrm{MAC}}} + \gamma^2\underbrace{\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_q\big[\log q_{\boldsymbol{\zeta}_{\boldsymbol{\theta}}}(\mathbf{z}|\mathbf{m}_i)\big]}_{\mathcal{L}_{\mathrm{PDF}}}, \tag{5}$$

where $\mathbf{d}_{ih} = \log\mathbf{f}_i - \mathbf{b} - \log\hat{\mathbf{f}}_i^h$. For unit-norm mode shapes perturbed by Gaussian noise, $1-\mathrm{MAC}$ equals the squared noise component perpendicular to the shape, so the MAC term is the Gaussian log-likelihood of that noise and does not depend on the sign of the modes. With these weights and a uniform prior, Eq. (5) is the negative ELBO and its minimizer is the Bayesian posterior when $\gamma = 1$; other values of γ produce tempered posteriors and are selected

on the validation set. We compute the PDF term through the entropy of the copula model,

$$\mathbb{E}_q[\log q] = -\tfrac{1}{2}\log\det\boldsymbol{\Sigma} - \sum_{e=1}^{7} H[q_e], \tag{6}$$

where the copula part is exact and the entropies of the marginals q_e are estimated with the H samples. This reduces the variance of the estimate compared with evaluating $\log q$ at the samples. The penalty that pushed samples away from the bounds and the term that forced diverse mixture weights were removed, since both shifted the optimum away from the posterior.

Sampling layer and training

The Kumaraswamy-mixture marginals are sampled by inverting their cumulative distribution function at the uniform variables produced by the Gaussian copula. The gradients of the samples with respect to the encoder outputs are computed exactly through the implicit function theorem, without clipping; they agree with finite differences to a relative error of 7×10^{-6}. Table 7 lists the training specifications. During the first 150 epochs, the data terms are multiplied by a factor that grows from 0.01 to 1, so that the approximate posterior first covers the plausible region and then concentrates on it. The model with the lowest validation loss after this warm-up is retained.

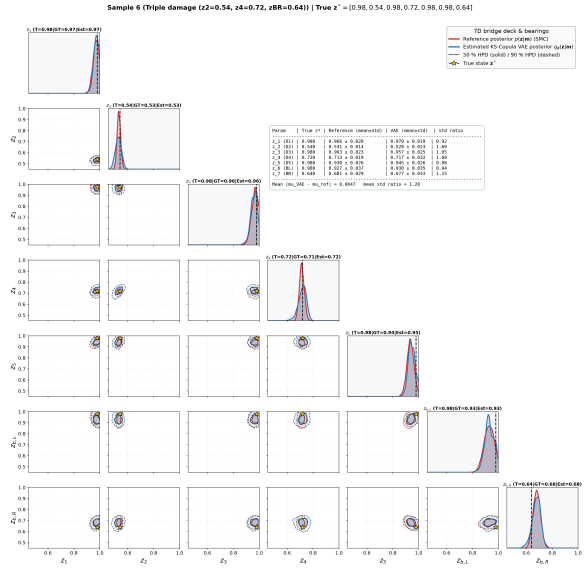
TABLE 7: Training specifications.

Encoder	three hidden layers (128, 256 and 128 neurons), ReLU, He uniform initialization
Approximate posterior	Gaussian copula with Kumaraswamy-mixture marginals ($K = 3$) on $[0.45, 1]$
Samples per observation	$H = 6$
Optimizer	Adam, cosine learning rate from 5×10^{-4} to 10^{-5}, batch size 256
Warm-up	data terms multiplied by a factor growing from 0.01 to 1 over the first 150 epochs
Epochs and model selection	1 500 epochs; lowest validation loss after the warm-up
Hyperparameter γ	$\gamma = 1$ (Bayesian posterior)
Hardware	16-thread CPU, 3–4 s per epoch

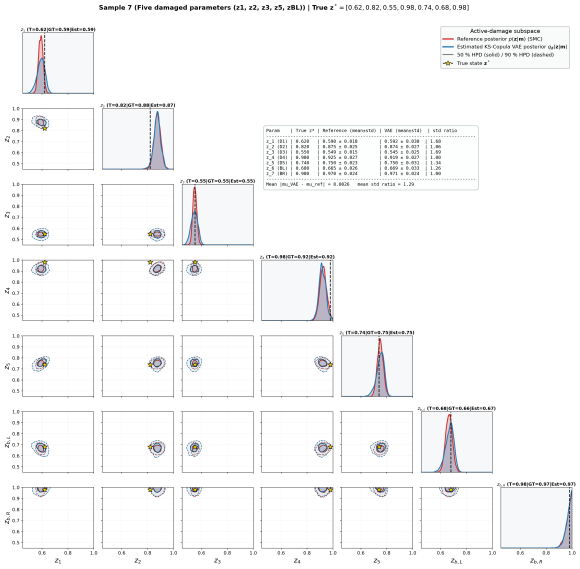
Verification of the corrected code

We first verified the corrected code on the intermediate dataset (five-element model, frequencies and both vertical and rotational mode shapes), whose exact posteriors are narrow and therefore demanding in terms of accuracy. After 1 500 epochs, the VAE reproduces the exact posteriors of the eleven benchmark scenarios (Fig. 6). The median ratio between the standard deviations

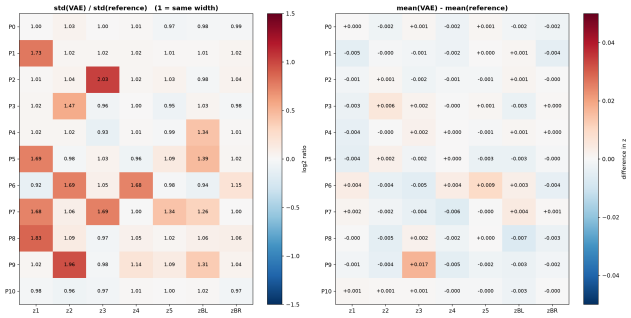
of the VAE and of the exact posterior is 1.02, the mean difference between their means is 0.002, and the true damage condition is contained in the 95% credible intervals of 94% (zones) and 93% (bearings) of the test parameters, compared with 72% for the previous implementation. Figure 7 shows the evolution of the loss and of each of its terms.



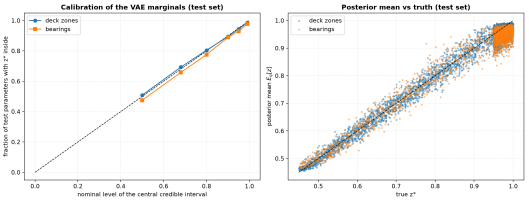
(a) Benchmark 6



(b) Benchmark 7 (damaged parameters)



(c) Ratio of standard deviations and difference of means



(d) Coverage and posterior means over the test set

FIG. 6: Verification of the corrected code on the intermediate dataset: exact posterior (red) and VAE posterior (blue); solid and dashed contours enclose 50% and 90% of the probability.

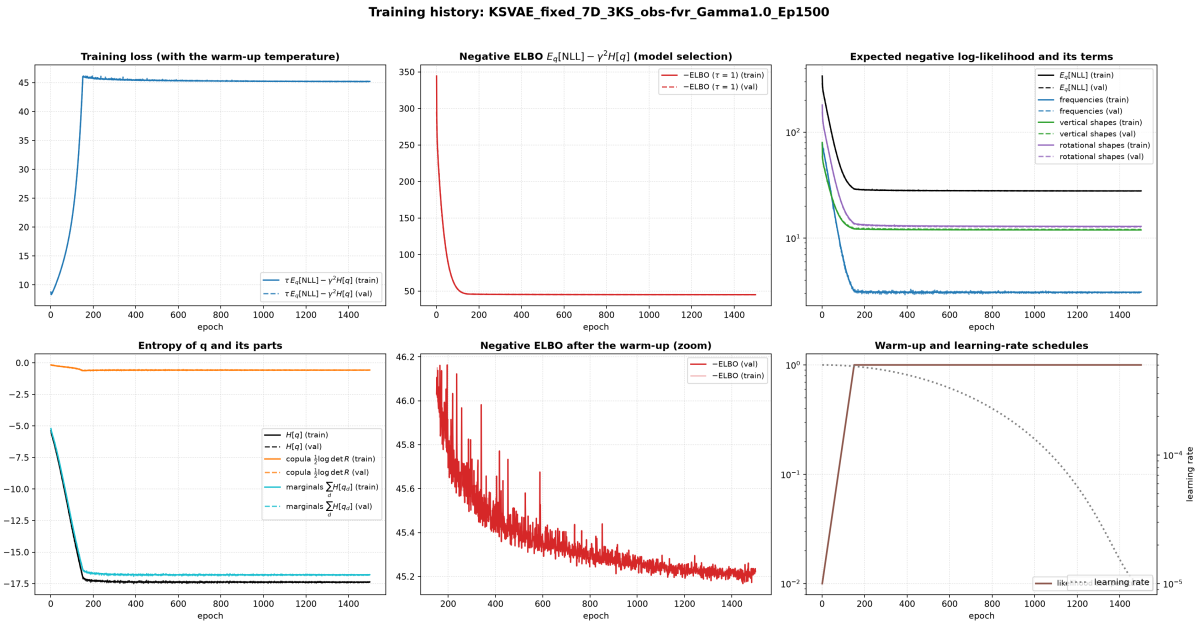


FIG. 7: Evolution of the loss and its terms for training and validation data.

The results of the VAE trained on the final dataset (Fig. 4) will be added when the corresponding training finishes.

POST-PROCESSING AND REFERENCE POSTERIOR

Exact posterior of the test scenarios

To assess the VAE, we need the exact posterior of each benchmark observation $\mathbf{m}_t$ under the same model that the VAE targets: the same decoder, the same calibrated likelihood and the same prior,

$$p(\mathbf{z}|\mathbf{m}_t) \propto \exp\big(-\mathrm{NLL}(\mathbf{z};\mathbf{m}_t)\big)\,\mathcal{U}(\mathbf{z};\mathcal{S}), \tag{7}$$

where NLL is the data part of Eq. (5) for a single sample. We draw 4 000 samples from this distribution with sequential Monte Carlo (SMC), which does not use the true damage condition at any stage. The algorithm starts from samples of the prior and introduces the likelihood gradually through an exponent β that grows from 0 to 1. At each step, β is increased as much as possible while keeping half of the samples effective, the samples are resampled according to their weights, and each of them is moved with random-walk Metropolis steps that leave the current tempered distribution invariant. Each benchmark requires 7–10 steps and 10–20 seconds on a CPU. Two independent runs give the same posterior (Wasserstein distance below 0.0013 per parameter), and a brute-force importance sampling with two million prior samples reproduces the SMC means and standard deviations within 0.002.

Comparison between estimated and exact posteriors

The VAE posterior of a test observation is represented by 4 000 samples drawn from $q_{\zeta_\theta}(\mathbf{z}|\mathbf{m}_t)$, without any further processing. Both posteriors are displayed with the same kernel density estimator, which accounts for the bounds of $\mathcal{S}$, and the joint densities are shown through the contours that enclose 50% and 90% of the probability. For each benchmark, we compare the means, the standard deviations and the correlations of both posteriors and compute the Wasserstein distance between their marginals. Over the whole test set, we compute the central credible intervals of the exact VAE marginals at nominal levels between 50% and 99%, their coverage of the true damage condition, their width, the error of the posterior mean, and the false-alarm rate on undamaged parameters.

Procedure

All steps are run from the project root (see `README.md`):

1. `dataset_generation.py` generates the datasets, calibrates the likelihood and plots the data figures.

2. `check_consistency.py` runs numerical checks of the likelihood, the sampling layer and the calibration.

3. `reference_posterior.py` computes and stores the exact posteriors of the benchmark scenarios.

4. `main_train_KSVAE.py` trains the VAE and stores the posterior parameters of the test set.

5. `postprocessing.py` produces the loss curves, the coverage analysis, the corner plots and the metrics.